



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CODEIGNITER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on MVC, routing, Query Builder, validation, and testing

TOTAL: 10 QUESTIONS

Q1 What is CodeIgniter and how does it implement MVC? [Threat Model](#)

Q6 How does CodeIgniter 4 handle sessions differently from PHP native sessions? [Security](#)

Q2 How does CodeIgniter 4 routing work? [Architecture](#)

Q7 How do you handle file uploads in CodeIgniter 4? [Lifecycle](#)

Q3 How does the CodeIgniter Query Builder prevent SQL injection? [Configuration](#)

Q8 What is a CodeIgniter RESTful Resource Controller? [Compliance](#)

Q4 What are CodeIgniter 4 Models and Entity classes? [Hardening](#)

Q9 How does CodeIgniter 4 Filters work? [Testing](#)

Q5 How does CodeIgniter validation work? [SSO / IdP](#)

Q10 How do you test a CodeIgniter 4 controller? [Tuning](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Codelgniter 4 is a lightweight PHP framework.

Mitigation

Codelgniter 4 is a lightweight PHP framework. Controllers in app/Controllers extend BaseController and handle HTTP logic. Models in app/Models extend Model and interact with the database. Views in app/Views are PHP files rendered with return view('name', data).

A6 CI4's session library wraps native sessions but

Observability

CI4's session library wraps native sessions but adds configurable drivers (files, database, Redis, Memcached). Sessions are signed with an encryption key to prevent tampering. set_flashdata('key', val) stores one-request messages. Regeneration is automated.

A2 Routes are defined in app/Config/Routes.php:

Primitives

Routes are defined in app/Config/Routes.php: \$routes->get('users/{:num}', 'UserController::show/\$1'). \$routes->resource('users') creates CRUD routes automatically. Auto-routing (deprecated) mapped URLs to controllers by convention.

A7 Access via \$file = \$this->request->getFile('avatar'). Call

Automation

Access via \$file = \$this->request->getFile('avatar'). Call \$file->isValid() and \$file->hasMoved() before moving. \$file->move(WRITEPATH.'uploads', \$file->getRandomName()) stores it. Validate MIME type with getClientMimeType() and check file size against maxSize.

A3 Query Builder methods accept raw values that

Hardening

Query Builder methods accept raw values that are automatically escaped and parameterized. \$db->where('id', \$id)->get('users') generates a prepared statement. Never concatenate user input into query strings; use Query Builder or \$db->query('SELECT * FROM u WHERE id=?', [\$id]).

A8 php artisan make:controller UserController --rest generates

Assessment

php artisan make:controller UserController --rest generates index, show, create, store, edit, update, delete stubs. \$routes->resource('users', ['controller'=>'UserController']) maps HTTP verbs to these methods. Override only the methods your resource needs.

A4 A Model extending CI4's Model gets CRUD

Gotchas

A Model extending CI4's Model gets CRUD helpers (find, insert, update, delete) and built-in validation. An Entity class maps a database row to a typed PHP object with casting (int, bool, datetime). Set protected \$returnType = UserEntity::class in the model.

A9 Filters (like middleware) run before or after

Assessment

Filters (like middleware) run before or after controllers. Define a filter class with before() and after() methods. Register in app/Config/Filters.php under \$aliases. Apply globally or per-route: \$routes->get('admin/*', 'AdminController', ['filter' => 'auth']).

A5 Use \$this->validate(\$rules) in a controller, where \$rules is an array like

Sederation

Use \$this->validate(\$rules) in a controller, where \$rules is an array like ['email' => 'required|valid_email']. Define reusable rule sets in app/Config/Validation.php. \$this->validator->getErrors() returns field-level error messages after a failed validation.

A10 Extend CIUnitTestCase. Use \$result =

Optimization

Extend CIUnitTestCase. Use \$result = \$this->withUri('http://example.com/users/1')->controller(UserController::class)->execute('show', 1). \$result->assertStatus(200) and \$result->assertSee('Alice') check the response without starting an HTTP server.